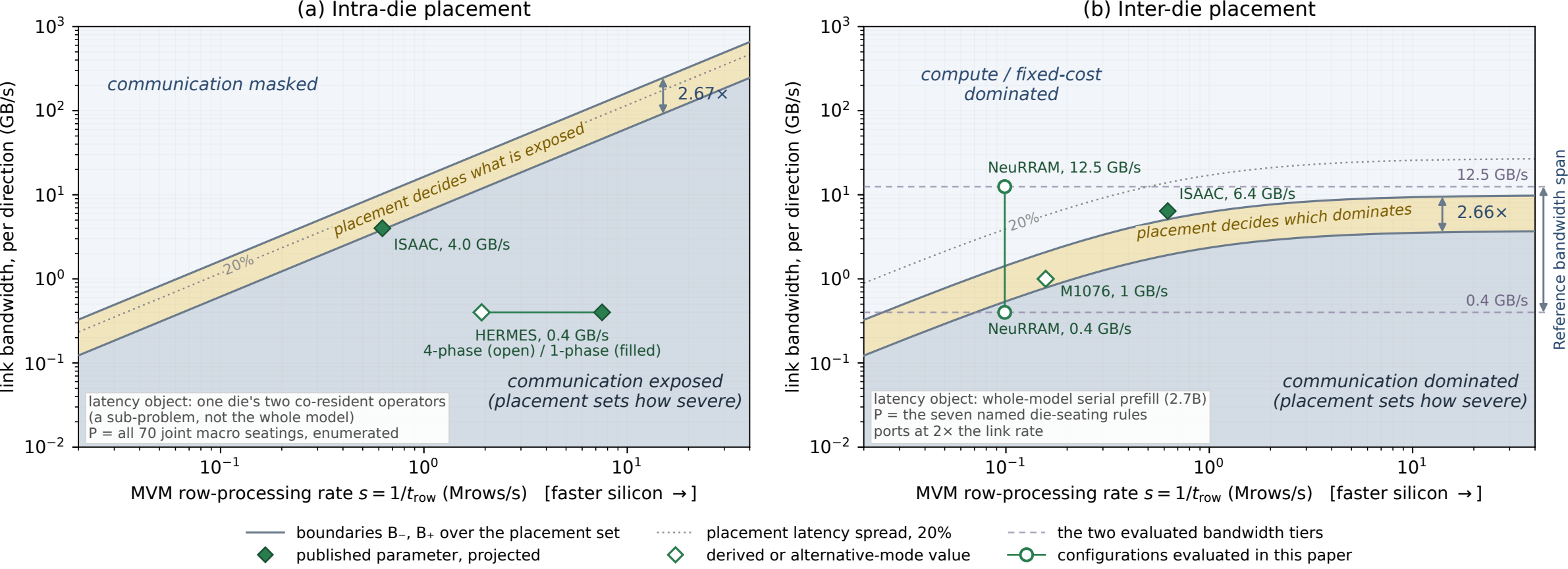
(a) Intra-die placement
(b) Inter-die placement
link bandwidth, per direction (GB/s)
MVM row-processing rate $s = 1/t_{row}$ (Mrows/s) [faster silicon →]
communication masked
placement decides what is exposed
20%
2.67×
ISAAC, 4.0 GB/s
HERMES, 0.4 GB/s
4-phase (open) / 1-phase (filled)
communication exposed
(placement sets how severe)
latency object: one die's two co-resident operators
(a sub-problem, not the whole model)
P = all 70 joint macro seatings, enumerated
compute / fixed-cost
dominated
NeuRRAM, 12.5 GB/s
12.5 GB/s
ISAAC, 6.4 GB/s
20%
placement decides which dominates
2.66×
M1076, 1 GB/s
0.4 GB/s
NeuRRAM, 0.4 GB/s
Reference bandwidth span
communication dominated
(placement sets how severe)
latency object: whole-model serial prefill (2.7B)
P = the seven named die-seating rules
ports at 2× the link rate
boundaries B–, B+ over the placement set
published parameter, projected
placement latency spread, 20%
derived or alternative-mode value
the two evaluated bandwidth tiers
configurations evaluated in this paper